

PORTABLE SOLAR USB POWER PORT

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A number of high-quality solar chargers are available in the market that come with adaptors to enable direct charging of handheld devices like mobile phones

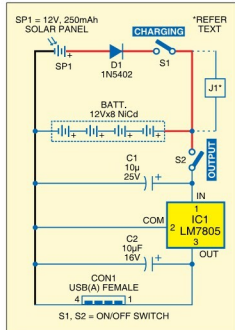


Fig. 1: Circuit of portable solar USB power port

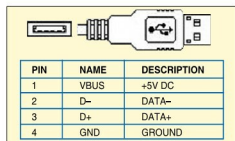


Fig. 2: USB connector and its pin configuration

and iPods. These chargers are supplied with a USB cable. One end of the cable is to be plugged into the device and the other end has a USB-A-type male plug for connection to the solar charger unit. The chargers consist of a small solar panel and an on-board rechargeable battery, which, in turn, is used to recharge the device (iPod, etc). The choice of the solar panel strongly depends on the capacity of the rechargeable battery used.

Fig. 1 shows an alternative circuit

for the portable solar USB power port. It consists of a solar panel (SP1), reflow-protection diode (D1), battery bank and fixed-voltage regulator 7805 (IC1).

The DC supply output from the solar panel (12V, 250mAh recommended)



Fig. 3: Enclosure for portable solar USB power port

is fed to the battery bank through the reflow-protection diode (1N5402). The battery bank contains eight 1.2V, 2700mAh rated, AA-type Ni-Cd cells connected in series.

As the rechargeable batteries used have a capacity of 2700 mAh, it is safe to use a charging current of 270 mA (10 per cent of 2700 mA). The nominal voltage of each battery is 1.2V, so the total voltage of eight batteries in series is 9.6V with a fully charged voltage of about 10.2V being normal. Therefore a 12V, 250mAh solar panel is recommended as a charger for the battery bank.

The output of the battery bank is connected to a fixed-voltage regulator circuit wired around IC LM7805. Steady 5V DC available at the output of IC1 is routed to a standard USB-A-type female plug.

Buy a cable with a moulded USB-A-type female connector and cut it off to use in this circuit. Connect the red USB cable lead to the 5V positive output



from IC1 and the black USB cable lead to the negative. The USB connector

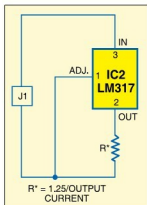


Fig. 4: Constant-current regulator add-on card



Fig. 5: Pin configuration of LM317

shown in Fig. 4. This add-on circuit is built around LM317, whose pin configuration is shown in Fig. 5. Use it only when you are using a differently rated solar panel and/or battery bank, as adequate current control may be necessary to prevent the cells from overcharging on very sunny days.

Connect the constant-current regulator circuit between switch S1 and the battery bank of the main circuit via two-pin jumper J1 after disconnecting the main circuit from switch S1. During testing, we had used a 5-ohm resistor (R). The value of R depends on the output current as follows: $R=1.25/\text{output current}$. ●